

5	(a) no hysteresis loop/no permanent deformation (do not allow 'force proportional to extension') so elastic change	M1 A0	
	(b) work done = area under graph line OR average force $\times$ distance $= \frac{1}{2}Fx$ $\frac{1}{2}(F_2 + F_1)(x_2 - x_1)$ $F = kx$, so work done = $= \frac{1}{2}kx^2$ $\frac{1}{2}k(x_2 + x_1)(x_2 - x_1)$ work done = $\frac{1}{2}k(x_2^2 - x_1^2)$	B1 A1 A1 A0	[3]
	(c) gain in energy of trolley = $\frac{1}{2}k(0.060^2 - 0.045^2) + \frac{1}{2}k(0.030^2 - 0.045^2)$ $= 0.36 \text{ J}$ kinetic energy = $\frac{1}{2} \times 0.85 \times v^2 = 0.36$ $v = 0.92 \text{ m s}^{-1}$	C1 C1 C1 A1	[4]

6	(a) (i) correct shape drawn	B1	[1]
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